DSAgent

Autonomous Pipeline Report

Dataset: titanic.csv
Generated: May 19, 2026 at 11:05 PM
Session: 13b1234f-d36…

19	6	38.7s
Steps Executed	Phases Completed	Total Time

1. Dataset Overview

The dataset contains **891** rows and **12** columns, using **0.28 MB** of memory.

Type	Count	Columns
Numeric	7	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Categorical	5	Name, Sex, Ticket, Cabin, Embarked

Numeric Statistics

Column	Mean	Median	Std	Min	Max
PassengerId	446.00	446.00	257.35	1.00	891.00
Survived	0.38	0.00	0.49	0.00	1.00
Pclass	2.31	3.00	0.84	1.00	3.00
Age	29.70	28.00	14.53	0.42	80.00
SibSp	0.52	0.00	1.10	0.00	8.00
Parch	0.38	0.00	0.81	0.00	6.00
Fare	32.20	14.45	49.69	0.00	512.33

AI Analysis

Key Insights about the Dataset • Dataset Overview: The dataset contains 891 rows and 12 columns, with a small memory footprint of 0.28 MB. • Data Types: The dataset includes 5 numeric columns, 5 categorical columns, and no datetime columns. • Survival Rate: The Survived column has a mean of 0.38, indicating that only about 38% of passengers survived. • Passenger Class: The majority of passengers were in Pclass 3 (mean = 2.31), with a significant number in Pclass 1 and Pclass 2. • Age Distribution: The average age is 29.7 years, with a wide range from 0.42 to 80 years. • Family Members: Most passengers traveled alone (mean SibSp = 0.52, mean Parch = 0.38). • Fare: The average fare is $32.2, with a large range from $0 to $512.33. Data Quality Issues Found • High Missing Values: • Cabin has 77.1% missing values (687 missing entries), making it potentially uninformative. • Age has 19.87% missing values (177 missing entries), which may affect model performance if not handled. • Embarked has 0.22% missing values (2 missing entries), which is negligible. • High Cardinality: • Name has 891 unique values, making it unsuitable for direct modeling without further processing. • No Duplicates: • The dataset has 0 duplicate rows, which is good for data integrity. Most Interesting/Useful Columns • Survived: The target variable for any predictive modeling task. • Pclass: Strong correlation with survival (negative correlation), suggesting class might be a key factor. • Age: Moderate correlation with survival and Pclass, indicating age could be a useful feature. • Sex: A strong categorical variable that likely has a significant impact on survival. • Fare: Weak but positive correlation with survival, suggesting that higher fares might correlate with better survival. • Embarked: While missing values are low, it may provide geographic or socio-economic context. Likely Target Variable (for ML) • Target Variable: Survived (binary classification: 0 = died, 1 = survived). • Reasoning: The Survived column is a clear binary outcome, and its correlation with other features (like Pclass, Age, and Fare) suggests it is a suitable target for a classification task (e.g., predicting survival based on passenger characteristics).

2. Data Quality Assessment

Found **3** columns with missing values out of 891 rows.

Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

Potential Issues:

• High cardinality categorical columns: ['Name']

3. Data Cleaning Actions

Age has 19.87% missing values

```
column: Age
strategy: median
action: Filled with median: 28.00
nulls_before: 177
nulls_after: 0
rows_after: 891
```

Embarked has 0.22% missing values

```
column: Embarked
strategy: mode
action: Filled with mode: S
nulls_before: 2
nulls_after: 0
rows_after: 891
```

Fare has a large range and potential outliers

```
column: Fare
method: iqr
threshold: 1.5
rows_removed: 116
rows_remaining: 775
removal_percentage: 13.02
```

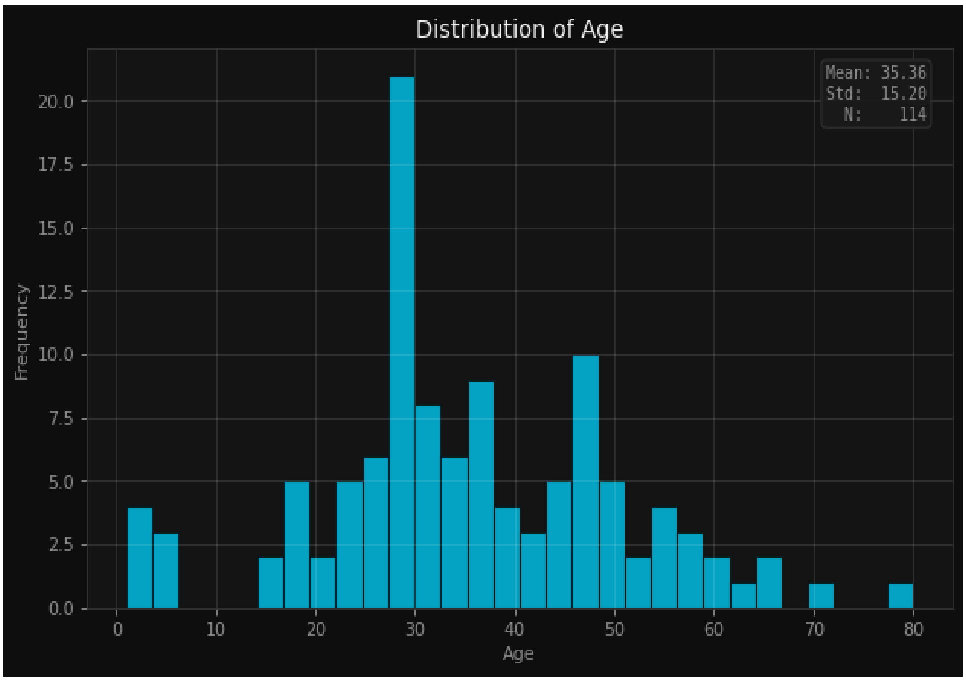
Cabin has 77.1% missing values and is potentially uninformative

```
column: Cabin
strategy: drop
action: Dropped 661 rows
nulls_before: 661
nulls_after: 0
rows_after: 114
```

Applied 4 cleaning operations based on data quality analysis.

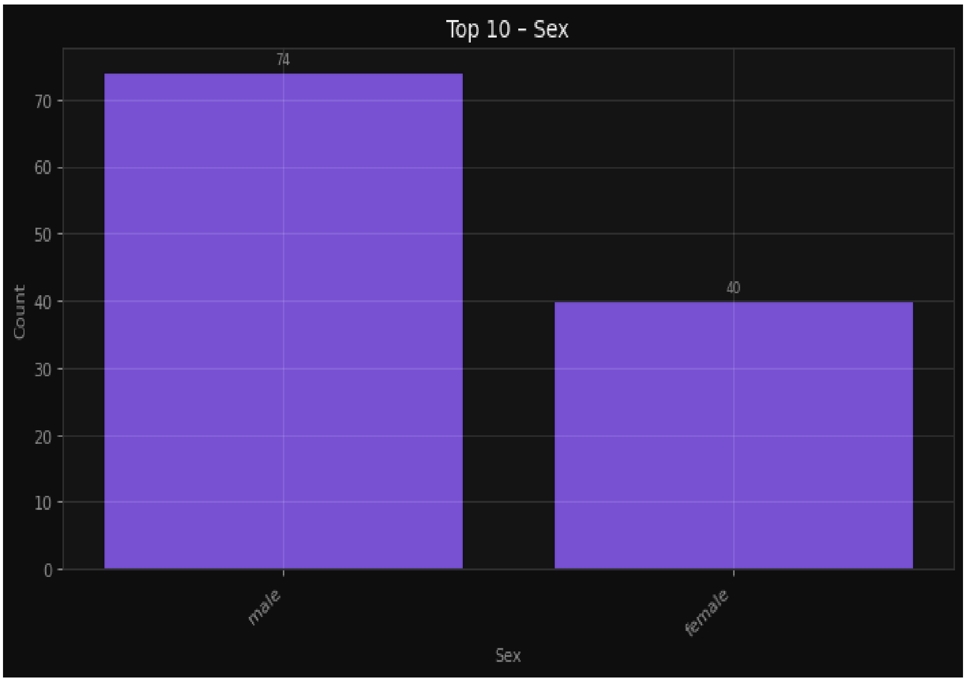
4. Visualizations & Insights

Understand the age distribution of passengers



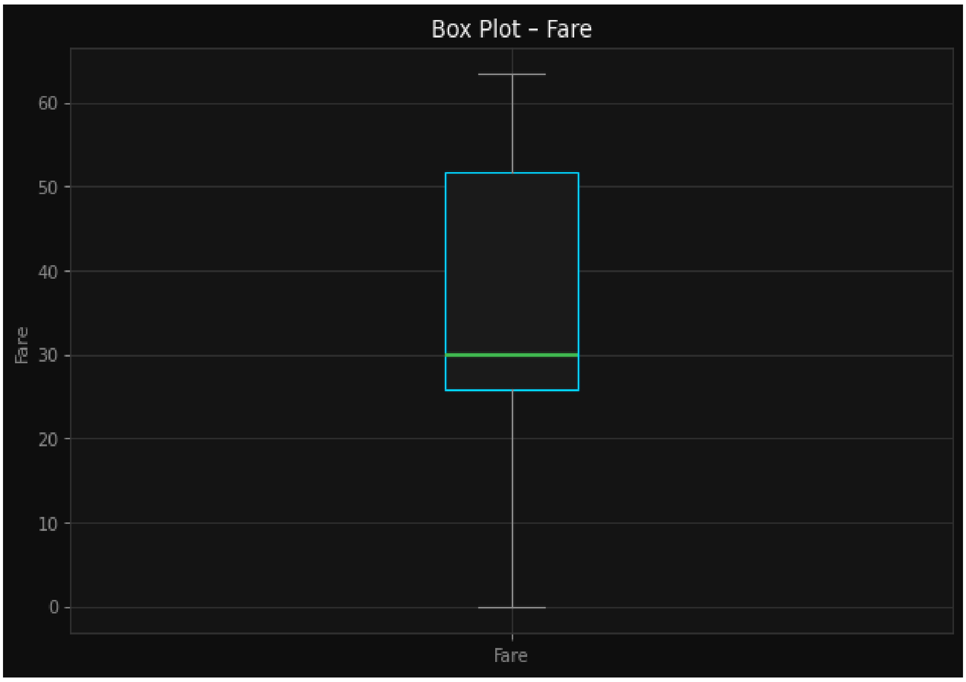
Understand the age distribution of passengers

Analyze the count of male and female passengers



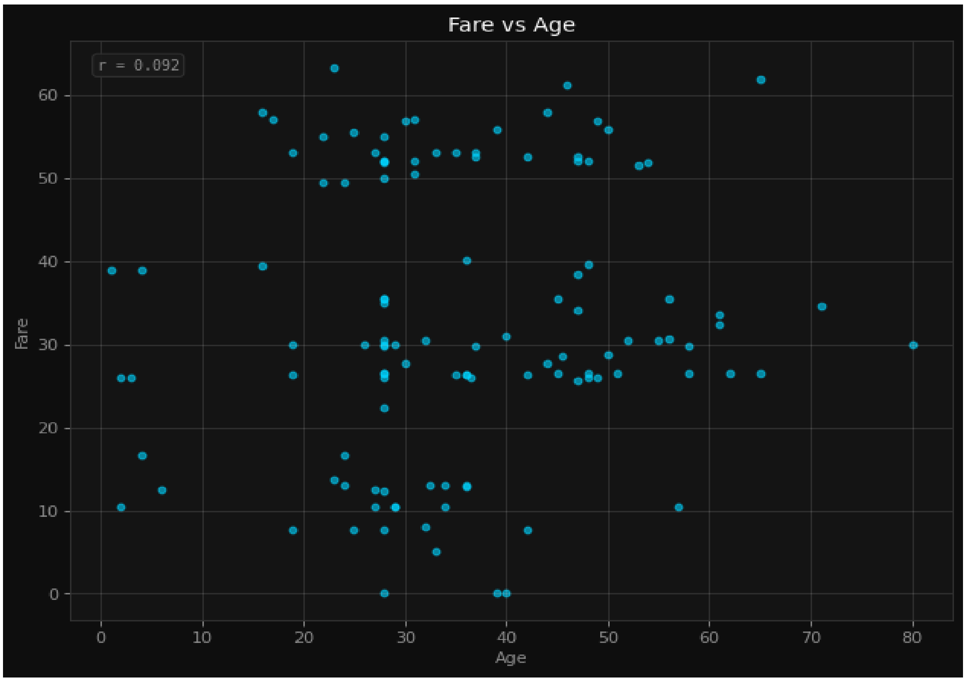
Analyze the count of male and female passengers

Detect outliers in fare payments



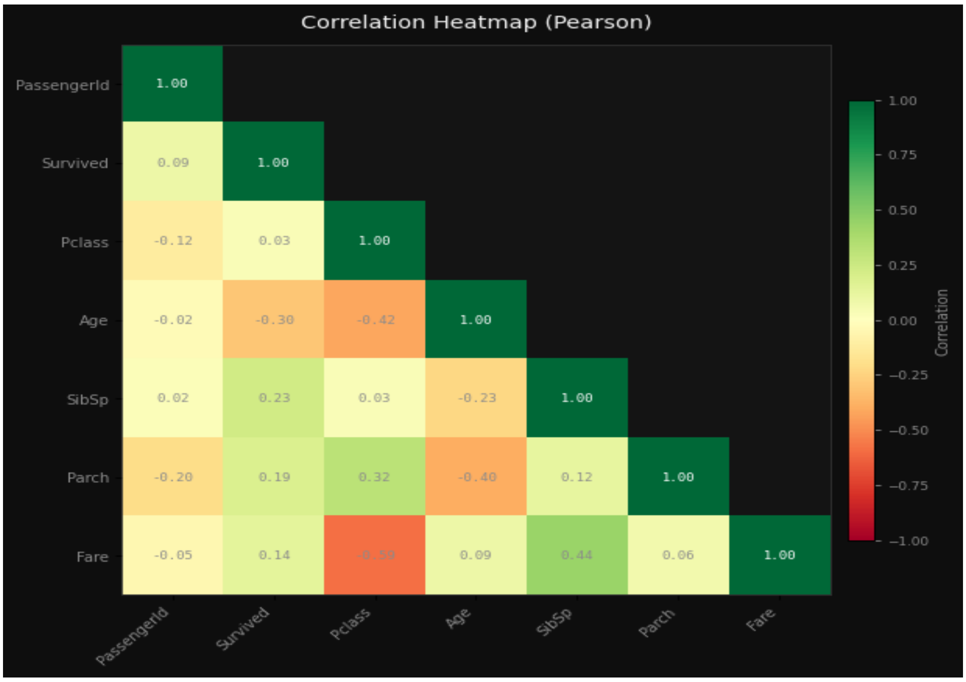
Detect outliers in fare payments

Explore the relationship between age and fare



Explore the relationship between age and fare

Identify correlations between numeric features



Identify correlations between numeric features

AI Interpretation

Histogram (Age Distribution): Likely shows a right-skewed distribution with a peak around 20–30 years, reflecting the typical age range of passengers on the Titanic. It may also highlight the presence of many missing values, especially in older age groups. Bar Chart (Male vs. Female Count): Likely reveals a significant majority of male passengers, consistent with historical context, as more men were on board compared to women. Box Plot (Fare Outliers): Likely shows a few high-fare outliers, possibly corresponding to first-class passengers, while the majority of fares fall within a narrower range. Scatter Plot (Age vs. Fare): Likely reveals a weak or no clear correlation, with some clustering of higher fares among older passengers, but no strong trend. Correlation Heatmap: Likely shows low or no correlation between most numeric features, with possibly a slight negative correlation between Age and Fare, and a strong correlation between Pclass and Fare (as lower Pclass often correlates with lower fare).

5. Feature Engineering

Encode categorical columns for ML modeling

```
encoding: One-Hot (get_dummies)
drop_first: True
columns_before: 12
columns_after: 13
new_columns_added: 1
rows: 114
```

Scale numeric features for better performance

```
scaler: StandardScaler (Z-score)
rows: 114
note: Dataset updated in session. Features now have μ≈0, σ≈1.
```

Reduce skew in Fare distribution

Applied 3 feature engineering steps.

6. Model Training & Comparison

Problem Type: **classification |** Target: **Embarked_S |** Features: **12**

Best Model: **XGBoost** — Score: **78.3%**

Model	Accuracy	Precision	Recall	F1
Random Forest	73.9%	69.4%	73.9%	68.8%
XGBoost	78.3%	76.5%	78.3%	75.5%
LightGBM	78.3%	76.5%	78.3%	75.5%
Logistic Regression	78.3%	76.5%	78.3%	75.5%

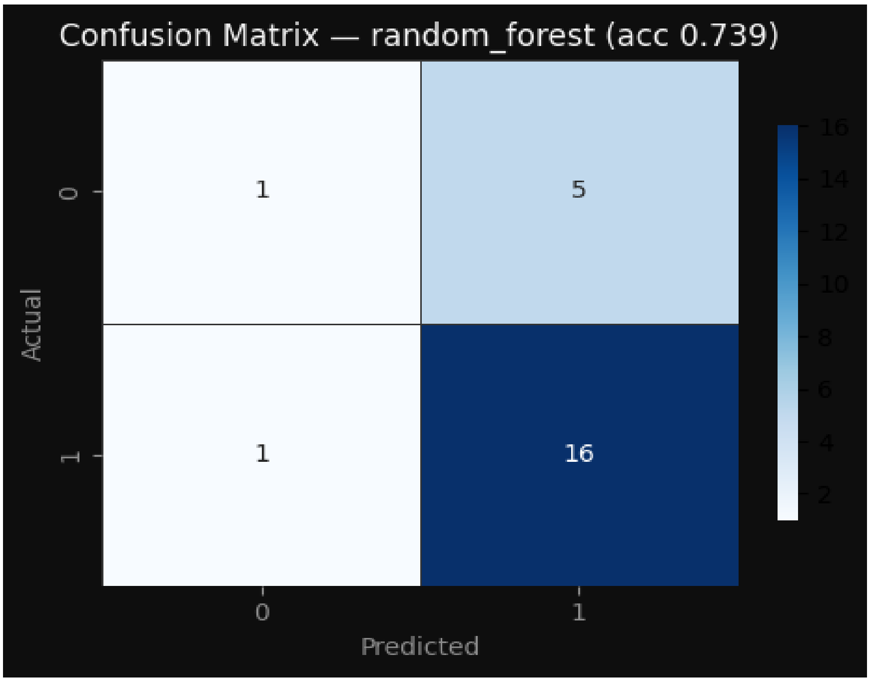
Model Selection Rationale

LLM error: 402

7. Model Evaluation

Model Evaluation Metrics

accuracy: 73.9% | precision: 69.4% | recall: 73.9% | f1_score: 68.8%



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

LLM error: 402